

COSMOTRON - HACKMITWPU'25

Cosmos Astronomy Club

Problem Statements

A	Observatory
A1	Create a system to optimize telescope observation schedules based on weather, object visibility, and scientific priorities.
A2	Design an improved star tracking system, that is less prone to errors.
A3	Create a system to automatically align observatory domes with telescope pointing directions to minimize obstruction.
A4	Design a camera system to automatically track and log satellites passing overhead.
A5	Use observatory data to generate maps of light pollution in real-time.
A6	Create a software tool that overlays images from radio, optical, and infrared telescopes.
A7	Create an algorithm to identify and remove cosmic ray artifacts from astronomical images in real time.
A8	Design a network of sensors to measure and map light pollution around observatories.
A9	Implement an AI-powered system to detect and alert operators to anomalies in observatory data and equipment.
A10	Develop a tool to seamlessly integrate and visualize data from multiple telescopes operating at different wavelengths.

B	Observational Astronomy
B1	Create a software pipeline to plot and analyze light curves for analyzing variable stars.
B2	Develop a system to identify unusual or unexpected patterns in large datasets, such as supernovae or gamma-ray bursts.
B3	Build an algorithm to identify gravitational lensing events in astronomical images.
B4	Design an AI model to detect transient astronomical events like supernovae or gamma-ray bursts in observational data
B5	Build an open-source web platform for amateur astronomers to contribute observational data.
B6	Develop a software pipeline for Calibration, Stacking and Reduction of astronomical images
B7	Modeling the Light Curve Evolution of GRBs: Implement an ML model to predict the light curve evolution of GRBs using observed early-time data, helping astronomers anticipate afterglow behavior.
B8	Create a machine learning model to classify galaxies based on their morphology from observational data
B9	Develop a predictive model for solar storm activity
B10	Noise Reduction in Stellar Flux Measurements: Use deep learning techniques (e.g., autoencoders, GANs) to denoise stellar flux data affected by instrumental noise, atmospheric interference, or cosmic rays.

C	Ground Station
C1	Design a cost-effective ground station system for educational and amateur satellite communication with frequency range from 30 MHz to 12 GHz
C2	Design a dashboard to visualize live telemetry and payload data from satellites.
C3	Create a system to process satellite telemetry data in real time for anomaly detection and diagnostics
C4	Design an optimal horn antenna with VSWR less than 1.2 with centre frequency at 1420 MHz for radio astronomy application
C5	Create an algorithm to automatically adjust ground station antenna positions for optimal signal reception.
C6	Create a tool to automatically correct Doppler shift in satellite signals for accurate data reception.
C7	Design a low cost omnidirectional antenna for Weather satellites with centre frequency at 1700 MHz
C8	Design a network of small antennas that work together to improve signal quality.
C9	Design a mobile ground station for rapid deployment in remote locations.
C10	Create a system to stream live ground station operations to schools and the public for educational purposes.

D	Satellite Optimization
D1	Design a compact imaging system for CubeSats to capture high-resolution Earth observation data.
D2	Design an optimization algorithm to minimize fuel usage during satellite orbital maneuvers
D3	Design a method for sensors to autonomously calibrate using celestial references (e.g., stars) to maintain accuracy over time.
D4	Design an alternative to patch antennas that maximizes power efficiency and signal quality while reducing weight and cost for satellite applications.
D5	Design an innovative magnetic torquer system to modify and stabilize the orbit of a satellite in low Earth orbit (LEO) for a space mission.
D6	Propose materials or coatings to reduce satellite reflectivity, mitigating light pollution for ground-based astronomy.
D7	Design a payload to monitor solar radiation, magnetic fields, and other space weather phenomena.

D8	Design a payload to test quantum communication protocols in space.
D9	Design a lossless compression algorithm tailored for hyperspectral imagery to reduce storage and transmission costs without compromising data quality.
D10	Design a payload to study the effects of microgravity on biological samples.

E	Space Debris Management
E1	Create a visualization tool to track and monitor space debris in real time using available data.
E2	Design an AI-driven system to predict potential collisions between operational satellites and space debris
E3	Develop software to analyze and predict debris density trends in various orbital regions
E4	Design a satellite constellation that minimizes the risk of collisions and debris creation.
E5	Build a simulation to test different strategies for satellite shielding against micrometeoroids and debris

F	Earth Observation
F1	Create a tool to monitor and analyze long-term climate change trends using Earth observation data.
F2	Develop an application that uses satellite imagery to track natural disasters like floods, wildfires, or earthquakes
F3	Create a system to analyze glacier retreat using satellite imagery and elevation data.
F4	Build an AI-based application to evaluate crop health and predict yields using multispectral satellite images
F5	Design a sensor to capture high-quality images of Earth at night.